

7(a)(i)	A photon is a <u>quantum of electromagnetic radiation with energy given by hf</u> , where h is the Planck's constant and f is the frequency of the wave.	[1] for correct answer
7(a)(ii)	The energy required by the hydrogen atom for it to transit from the ground state to its first excited state is 10.2 eV.	[1] for correct answer
7(b)(i)	$E_3 \rightarrow E_1$: $\Delta E = E_3 - E_1$ $= -9.0 \times 10^{-19} - (-24.6 \times 10^{-19})$ $= 15.6 \times 10^{-19} \text{ J}$ $\Delta E = hf$ $f = \frac{\Delta E}{h} = \frac{15.6 \times 10^{-19}}{6.63 \times 10^{-34}}$ $= 2.353 \times 10^{15}$ $\approx 2.35 \times 10^{15} \text{ Hz}$	 [1] for correct expression and numerical substitution [1] for correct answer
7(b)(ii)	$\Delta E_{31} = \Delta E_{32} + \Delta E_{21}$ $\Delta E_{21} = \Delta E_{31} - \Delta E_{32}$ $\frac{hc}{\lambda} = h(f_{31} - f_{32})$ $\lambda = \frac{hc}{h(f_{31} - f_{32})}$ $= \frac{3.00 \times 10^8}{2.35 \times 10^{15} - 1.09 \times 10^{15}}$ $= 2.381 \times 10^{-7}$ $\approx 238 \text{ nm}$	 [1] for correct method and numerical substitution [1] for correct answer
7(c)	The <u>accelerated electrons strike the anode metal and knock out electrons from the inner shells of the target atoms</u>, causing the electrons in the innermost shell to be vacant. An electron in an outer shell transits from a higher energy level to <u>fill up the vacancy in the inner shell</u>, and an X-ray photon can be produced. Since the energy transitions between discrete energy levels are fixed, the <u>energy gaps produce photons of fixed energies</u>.	 [1] [1] [1]

No.	Solution	Remarks
8(a)	Simple harmonic motion is an oscillatory motion in which the	[1] for a